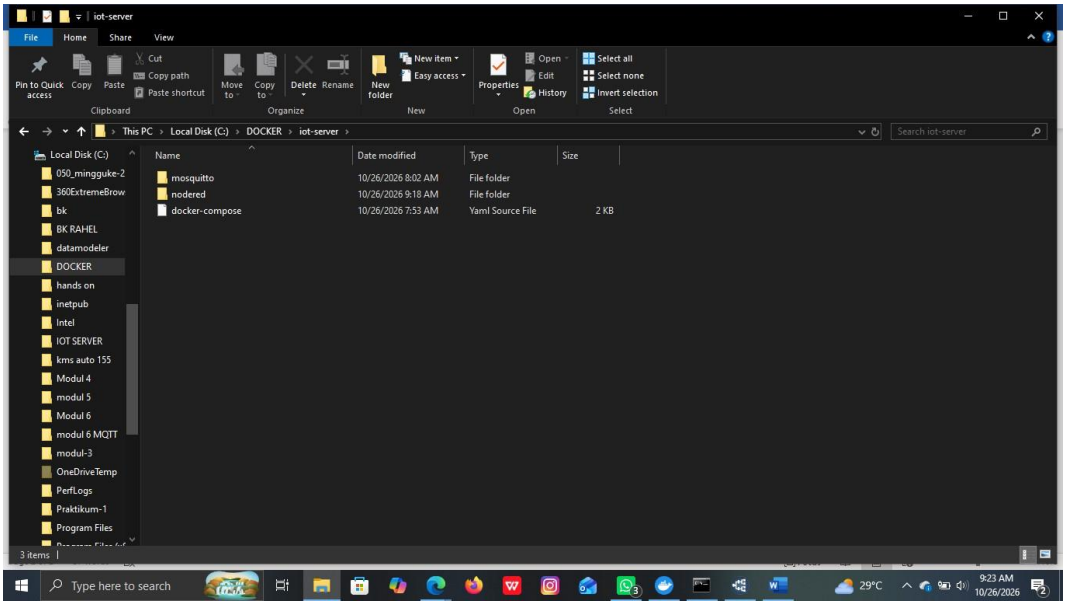
LAPORAN PRAKTIKUM
INTERNET OF THINGS

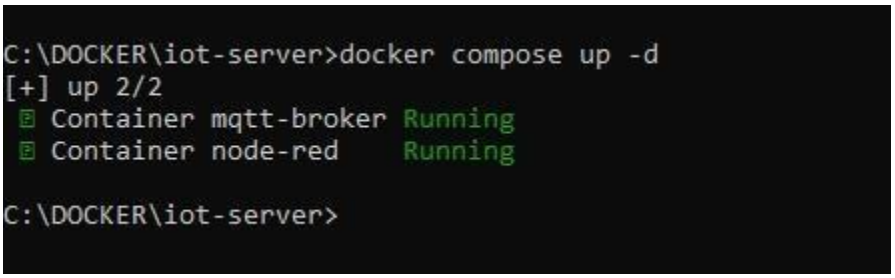
Dosen Pengampu: 1. Dr. Septafiansyah Dwi Putra S.T., M.T
2. Agus Ambarwari S.Pd., M.Kom

Disusun Oleh :
Nama : Rahel Nisa Septiyana NPM
: 25782050
Kelas : TRI 2B

PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK NEGERI LAMPUNG
TAHUN 2026

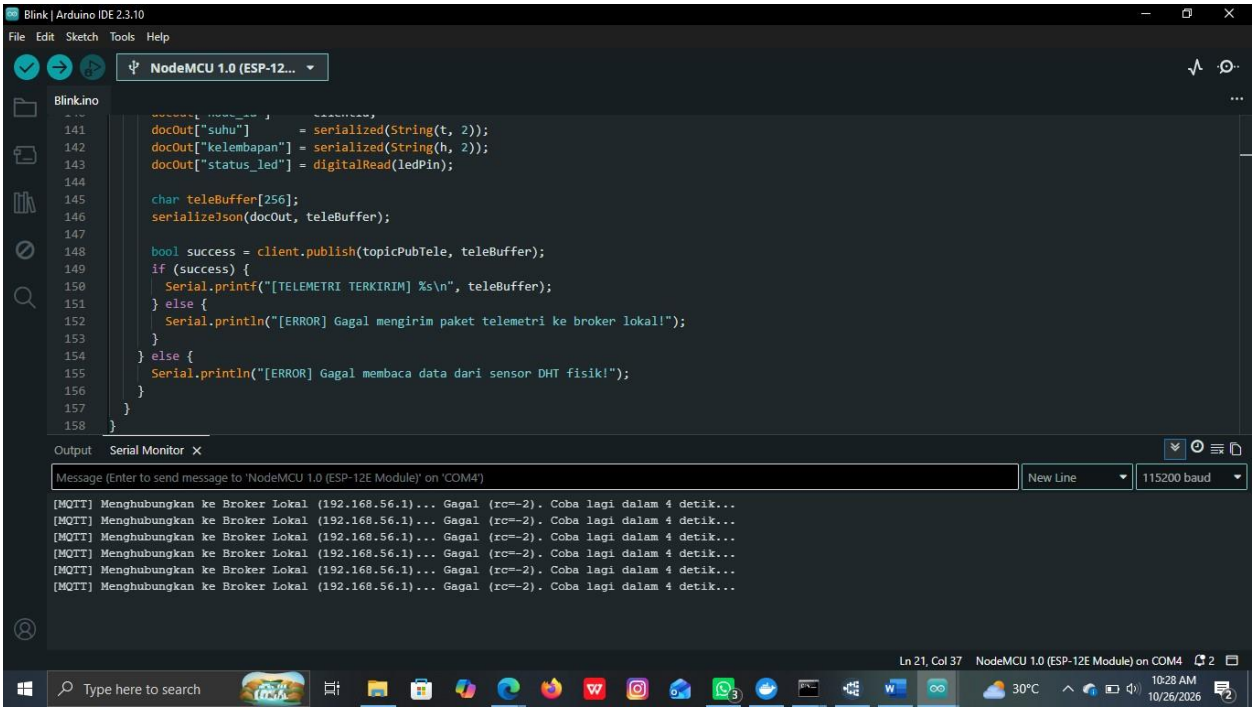
7.5.1 Praktikum 1: Pembangunan Ekosistem IoT Server dengan Docker Compose



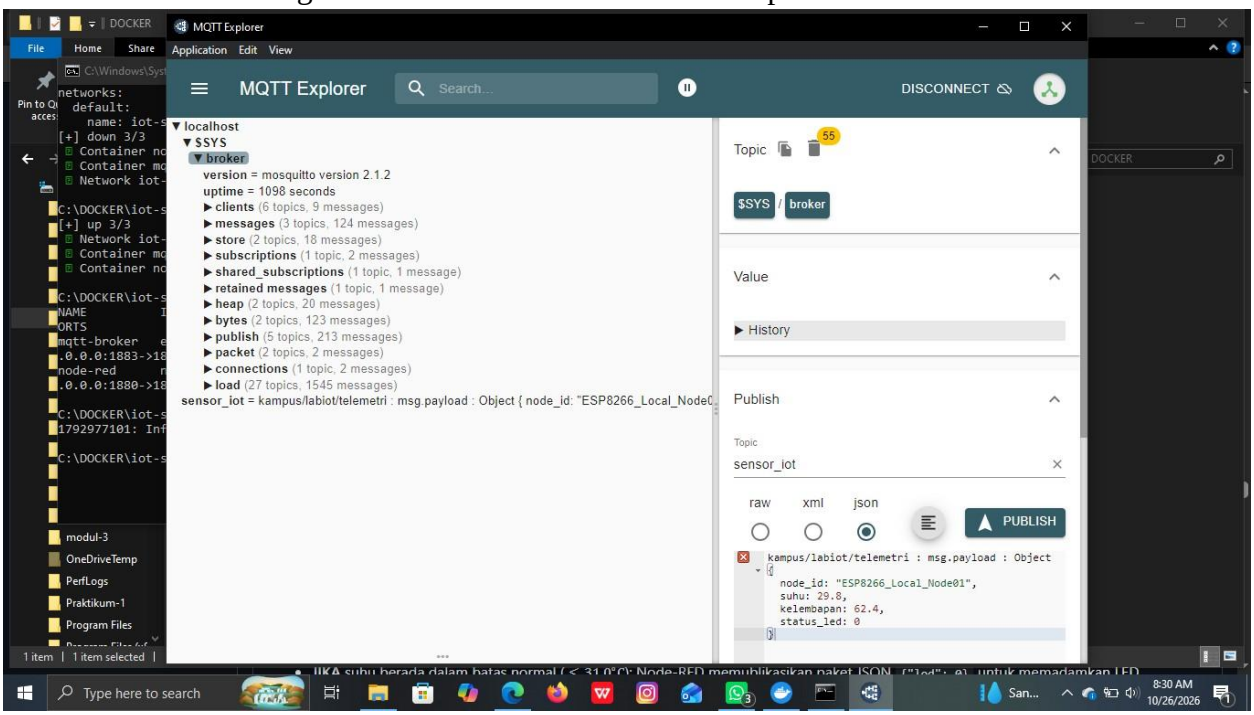


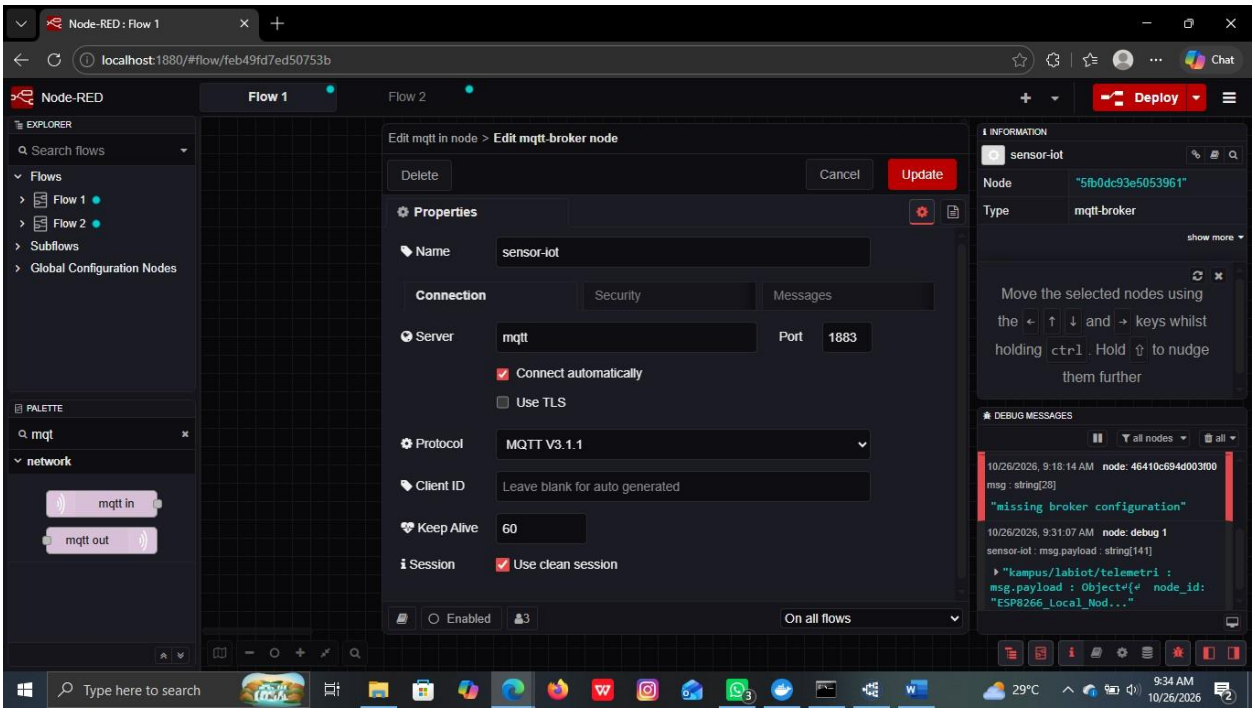


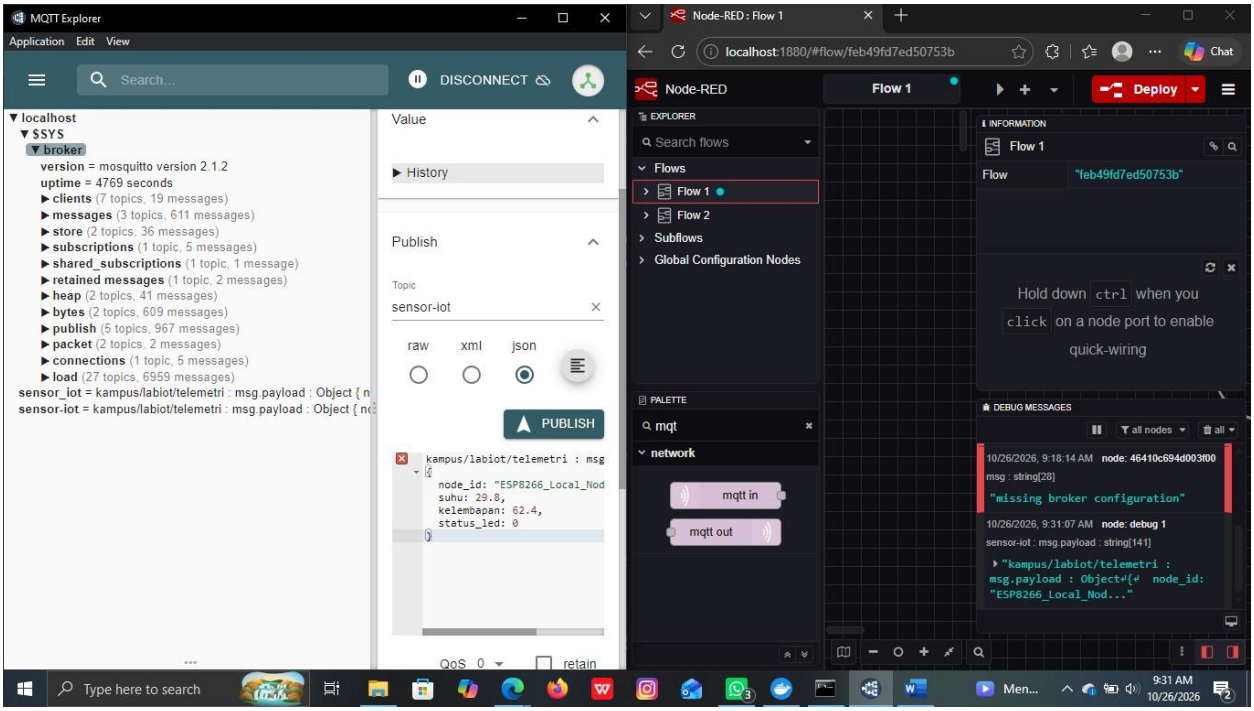
7.5.2 Praktikum 2: Migrasi Simpul NodeMCU ESP8266 ke Broker MQTT Lokal

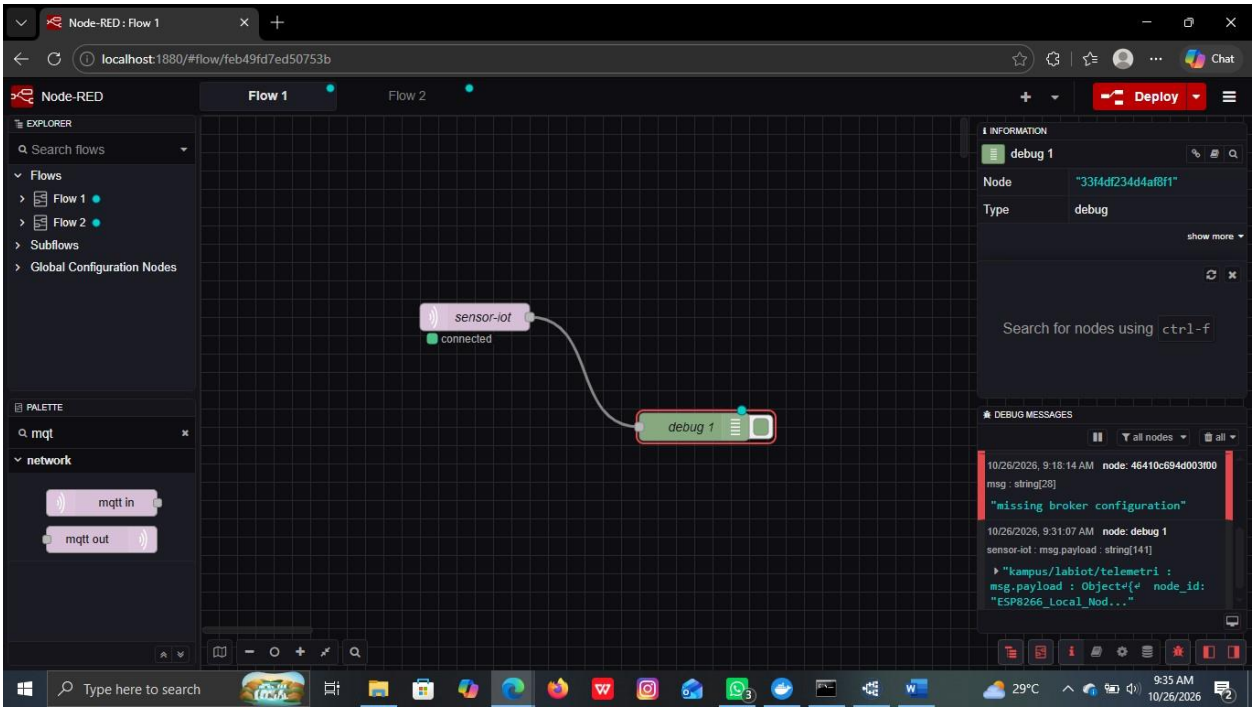

Blink | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
Blink.ino
docOut["suhu"] = serialized(String(t, 2));
docOut["kelembapan"] = serialized(String(h, 2));
docOut["status_led"] = digitalRead(ledPin);
char teleBuffer[256];
serializeJson(docOut, teleBuffer);
bool success = client.publish(topicPubTele, teleBuffer);
if (success) {
Serial.printf("[TELEMETRI TERKIRIM] %s\n", teleBuffer);
} else {
Serial.println("[ERROR] Gagal mengirim paket telemetri ke broker lokal!");
}
} else {
Serial.println("[ERROR] Gagal membaca data dari sensor DHT fisik!");
}
}
}
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM4')
New Line
115200 baud
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
Ln 21, Col 37 NodeMCU 1.0 (ESP-12E Module) on COM4
Type here to search
30°C
10:28 AM
10/26/2026

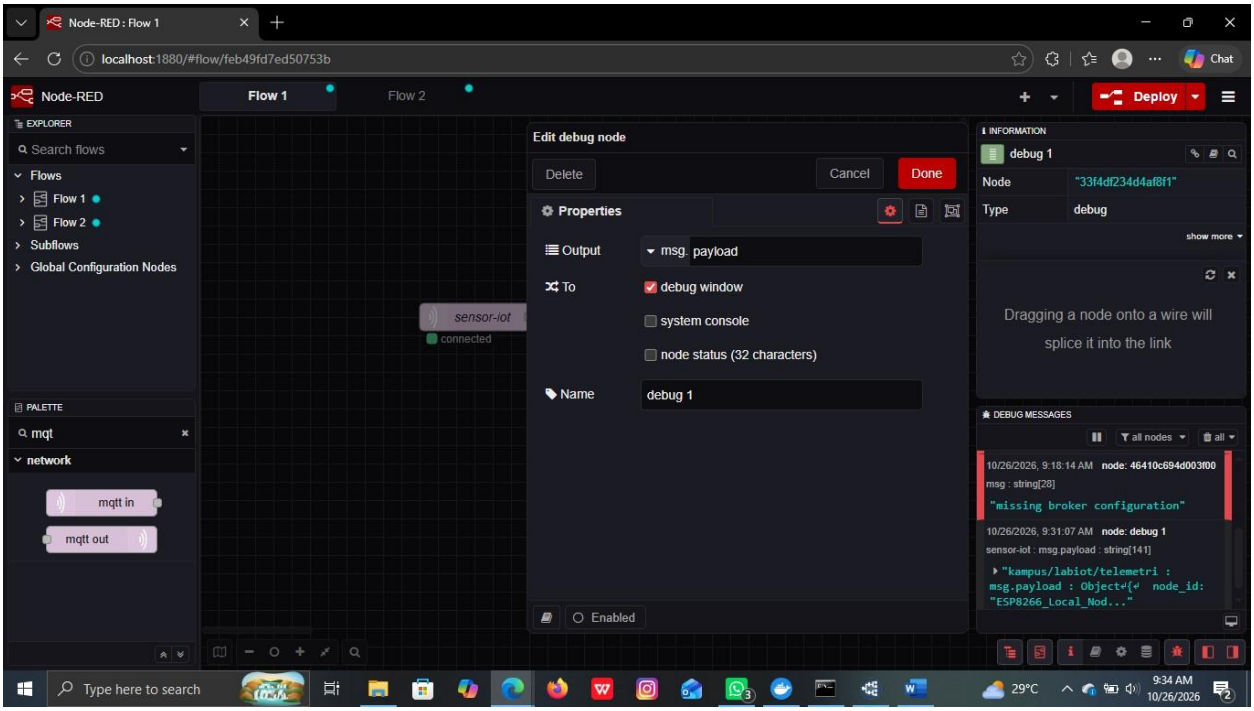
7.5.3 Praktikum 3: Integrasi Alur Data Telemetri Dua Arah pada Node-RED



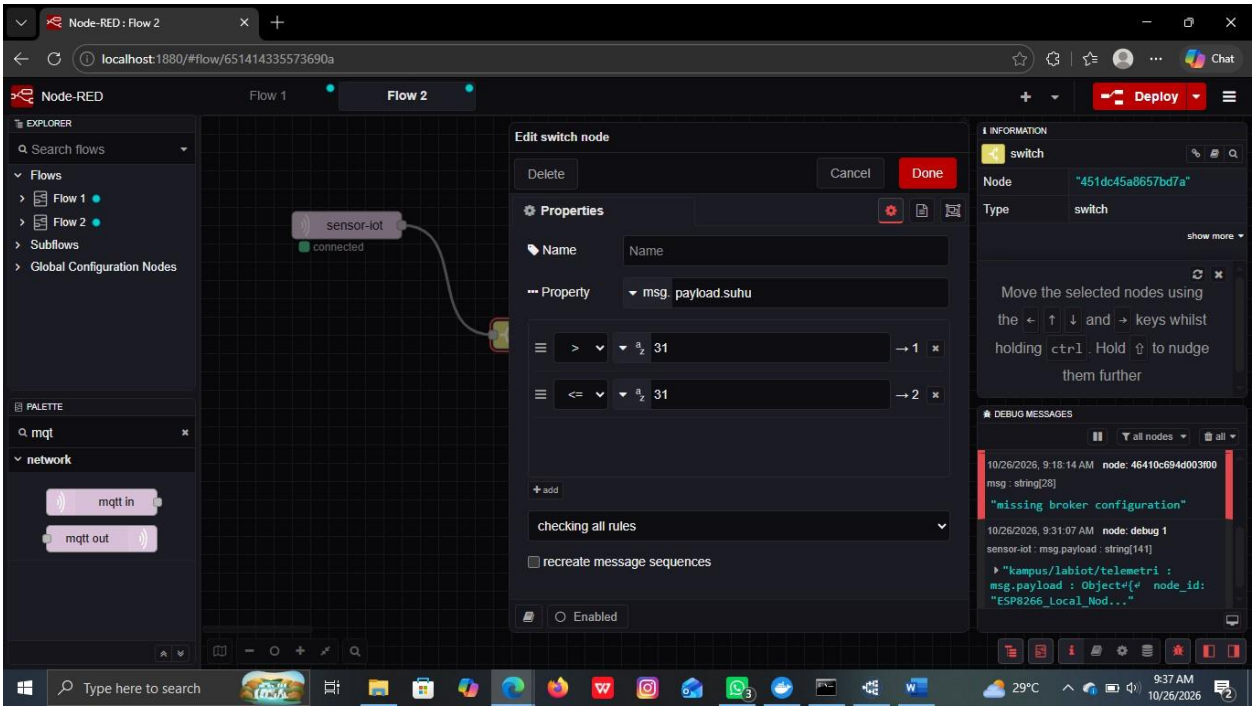


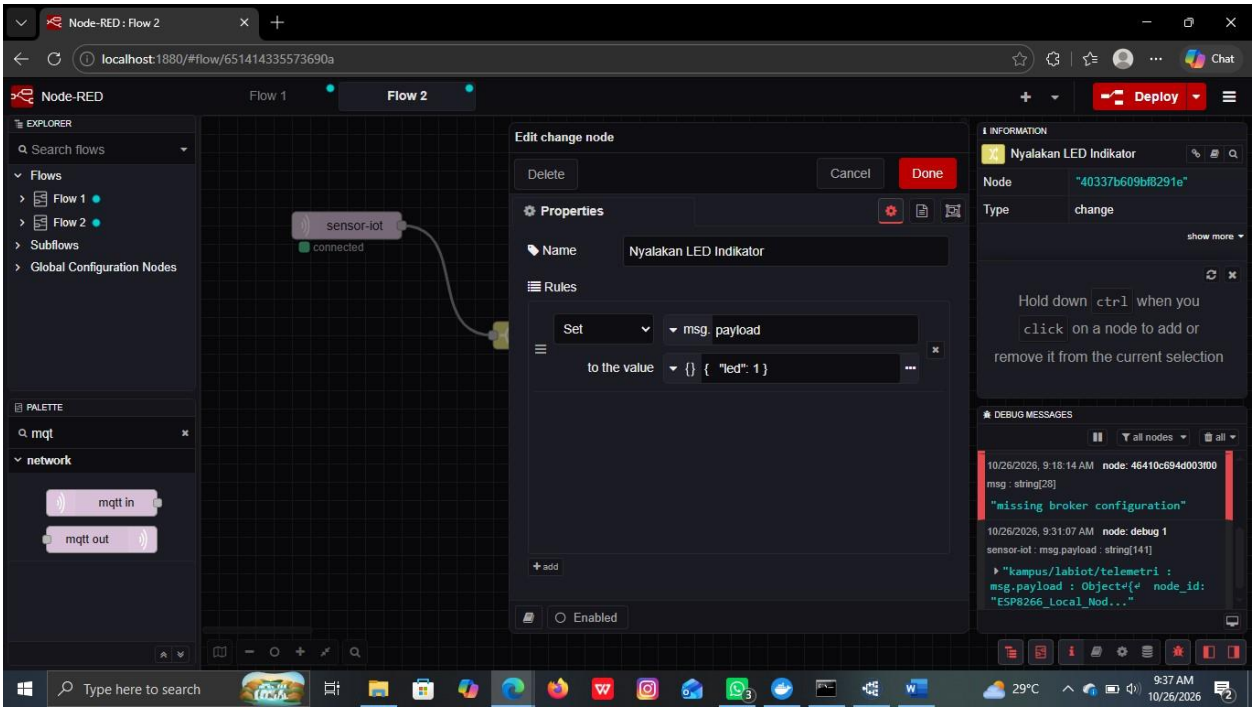

MQTT Explorer
Application Edit View
Search...
DISCONNECT
localhost
$SYS
broker
version = mosquitto version 2.1.2
uptime = 4769 seconds
clients (7 topics, 19 messages)
messages (3 topics, 611 messages)
store (2 topics, 36 messages)
subscriptions (1 topic, 5 messages)
shared_subscriptions (1 topic, 1 message)
retained messages (1 topic, 2 messages)
heap (2 topics, 41 messages)
bytes (2 topics, 609 messages)
publish (5 topics, 967 messages)
packet (2 topics, 2 messages)
connections (1 topic, 5 messages)
load (27 topics, 6959 messages)
sensor_iot = kampus/labiot/telemetri : msg.payload : Object { n
sensor-iot = kampus/labiot/telemetri : msg.payload : Object { n
Value
History
Publish
Topic
sensor-iot
raw
xml
json
PUBLISH
kampus/labiot/telemetri : msg
{
node_id: "ESP8266_Local_Nod
suhu: 29.8,
kelembapan: 62.4,
status_led: 0
}
QoS 0
retain
Node-RED : Flow 1
localhost:1880/#flow/feb49fd7ed50753b
Chat
Node-RED
Flow 1
Deploy
EXPLORER
Search flows
Flows
Flow 1
Flow 2
Subflows
Global Configuration Nodes
INFORMATION
Flow 1
Flow "feb49fd7ed50753b"
Hold down ctrl when you click on a node port to enable quick-wiring
PALETTE
mqt
network
mqtt in
mqtt out
DEBUG MESSAGES
all nodes
all
10/26/2026, 9:18:14 AM node: 46410c694d003f00
msg : string[28]
"missing broker configuration"
10/26/2026, 9:31:07 AM node: debug 1
sensor-iot : msg.payload : string[141]
"kampus/labiot/telemetri : msg.payload : Object↵{↵ node_id: "ESP8266_Local_Nod..."
Type here to search
Men...
9:31 AM
10/26/2026

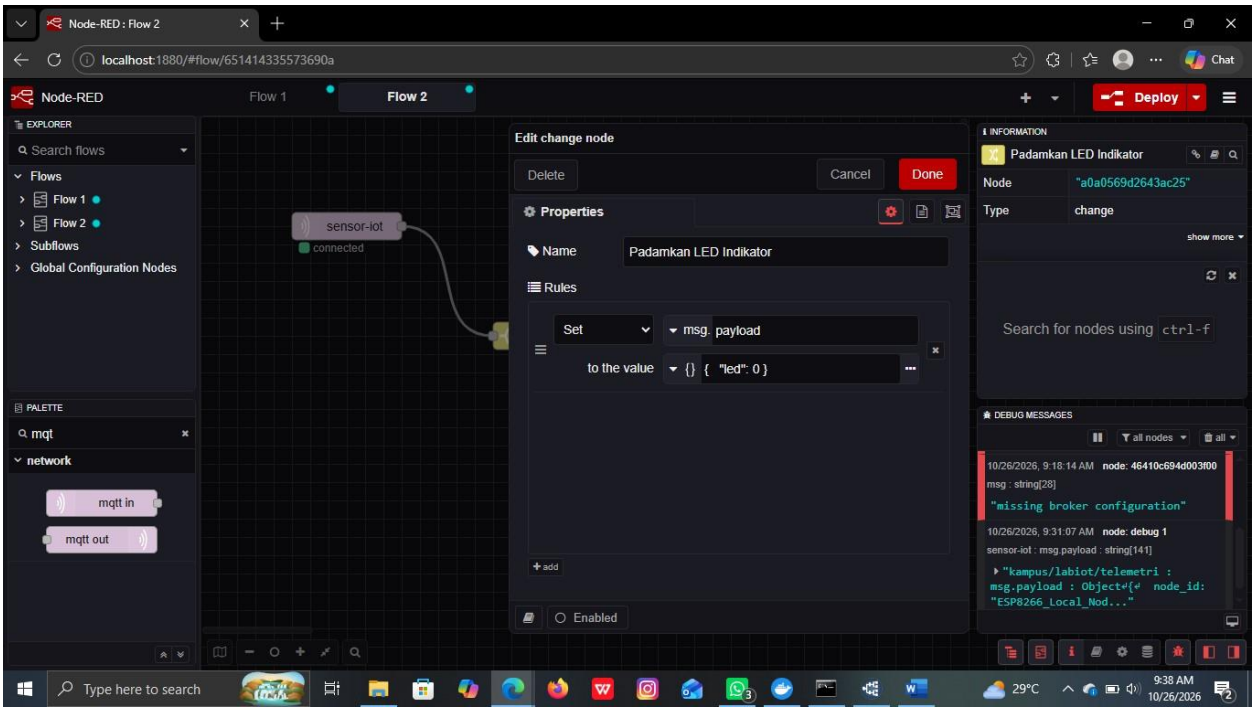

Node-RED : Flow 1
localhost:1880/#flow/feb49fd7ed50753b
Chat
Node-RED
Flow 1
Flow 2
Deploy
EXPLORER
Search flows
Flows
Flow 1
Flow 2
Subflows
Global Configuration Nodes
sensor-iot
connected
debug 1
INFORMATION
debug 1
Node "33f4df234d4af8f1"
Type debug
show more
Search for nodes using ctrl-f
PALETTE
mqt
network
mqtt in
mqtt out
DEBUG MESSAGES
all nodes
all
10/26/2026, 9:18:14 AM node: 46410c694d003f00
msg : string[28]
"missing broker configuration"
10/26/2026, 9:31:07 AM node: debug 1
sensor-iot : msg.payload : string[141]
"kampus/labiot/telemetri : msg.payload : Object↵{↵ node_id: "ESP8266_Local_Nod..."
Type here to search
29°C
9:35 AM
10/26/2026

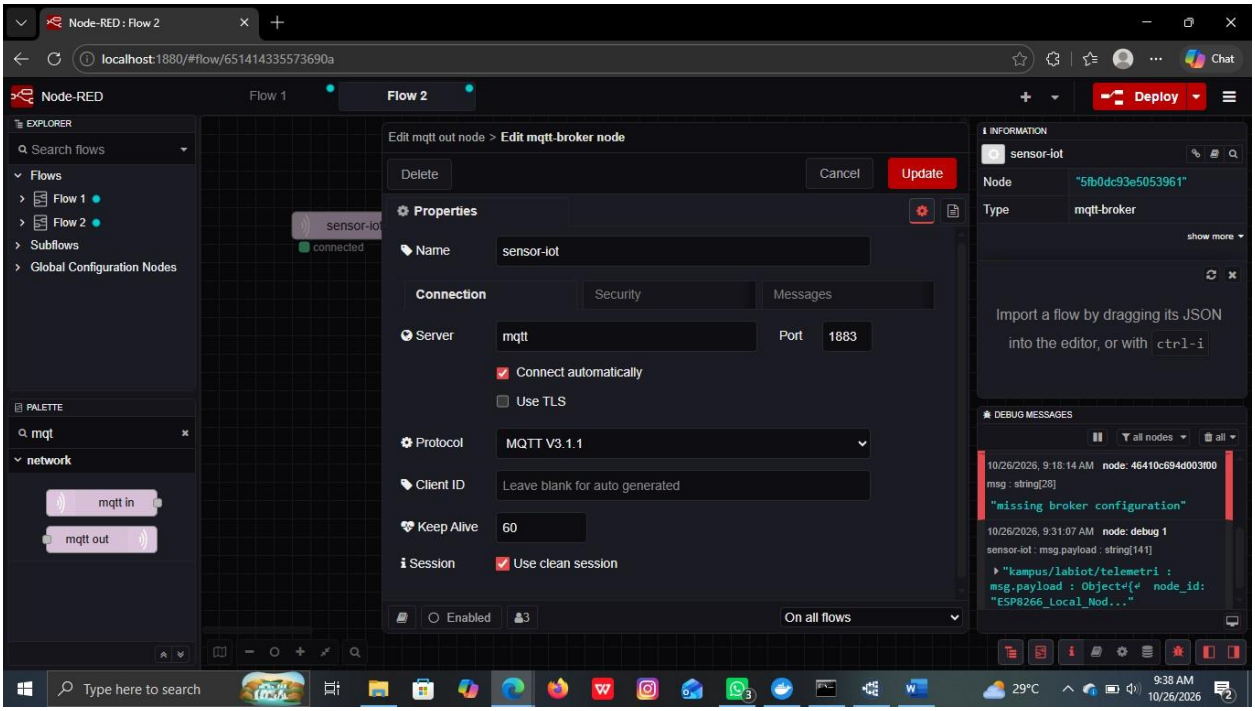


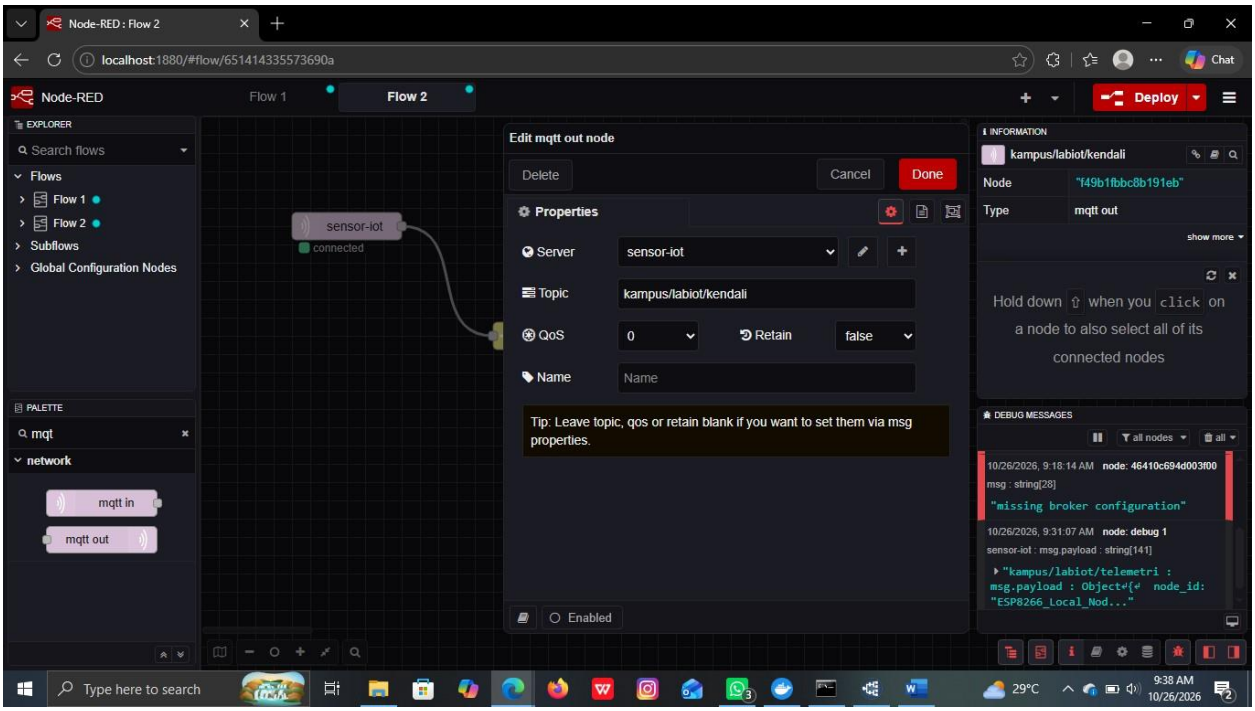
7.5.4 Praktikum 4: Implementasi Rule Engine Otomasi Aktuator Loop Tertutup

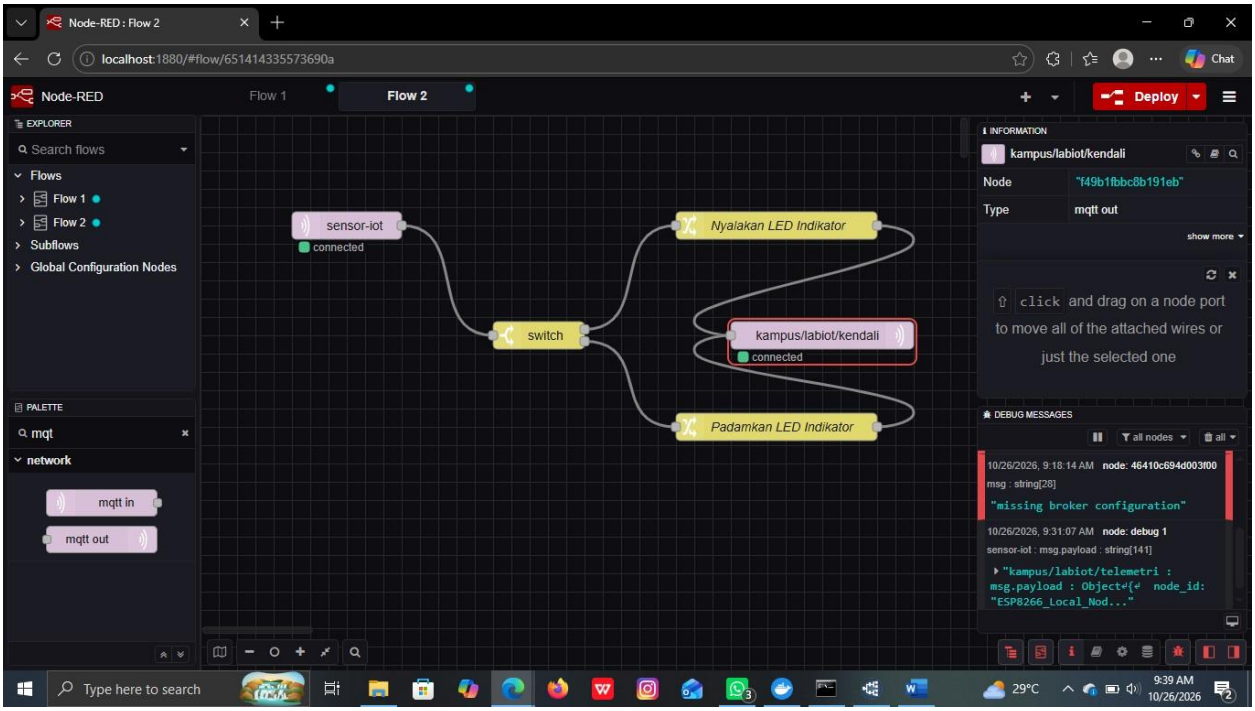



Node-RED : Flow 2
localhost:1880/#flow/651414335573690a
Node-RED
Flow 1
Flow 2
Deploy
EXPLORER
Search flows
Flows
Flow 1
Flow 2
Subflows
Global Configuration Nodes
sensor-iot
connected
Edit change node
Delete
Cancel
Done
Properties
Name
Nyalakan LED Indikator
Rules
Set
msg. payload
to the value
{ "led": 1 }
add
Enabled
INFORMATION
Nyalakan LED Indikator
Node
"40337b609bf8291e"
Type
change
show more
Hold down ctrl when you click on a node to add or remove it from the current selection
DEBUG MESSAGES
all nodes
10/26/2026, 9:18:14 AM node: 46410c694d003f00
msg : string[28]
"missing broker configuration"
10/26/2026, 9:31:07 AM node: debug 1
sensor-iot : msg.payload : string[141]
"kampus/labiot/telemetri : msg.payload : Object{ node_id: "ESP8266_Local_Nod..."
PALETTE
mqt
network
mqtt in
mqtt out
Type here to search
29°C
9:37 AM
10/26/2026


Node-RED : Flow 2
localhost:1880/#flow/651414335573690a
Node-RED
Flow 1
Flow 2
Deploy
EXPLORER
Search flows
Flows
Flow 1
Flow 2
Subflows
Global Configuration Nodes
sensor-iot
connected
Edit change node
Delete
Cancel
Done
Properties
Name
Padamkan LED Indikator
Rules
Set
msg. payload
to the value
{ "led": 0 }
add
Enabled
INFORMATION
Padamkan LED Indikator
Node
"a0a0569d2643ac25"
Type
change
show more
Search for nodes using ctrl-f
DEBUG MESSAGES
all nodes
10/26/2026, 9:18:14 AM node: 46410c694d003f00
msg : string[28]
"missing broker configuration"
10/26/2026, 9:31:07 AM node: debug 1
sensor-iot : msg.payload : string[141]
"kampus/labiot/telemetri : msg.payload : Object{ node_id: "ESP8266_Local_Nod..."
PALETTE
mqt
network
mqtt in
mqtt out
Type here to search
29°C
9:38 AM
10/26/2026


Node-RED : Flow 2
localhost:1880/#flow/651414335573690a
Node-RED
Flow 1
Flow 2
Deploy
Edit mqtt out node > Edit mqtt-broker node
Delete
Cancel
Update
Properties
Name
sensor-iot
Connection
Security
Messages
Server
mqtt
Port
1883
Connect automatically
Use TLS
Protocol
MQTT V3.1.1
Client ID
Leave blank for auto generated
Keep Alive
60
Session
Use clean session
Enabled
On all flows
INFORMATION
sensor-iot
Node
"5fb0dc93e5053961"
Type
mqtt-broker
show more
Import a flow by dragging its JSON into the editor, or with ctrl-i
DEBUG MESSAGES
10/26/2026, 9:18:14 AM node: 46410c694d003f00
msg : string[28]
"missing broker configuration"
10/26/2026, 9:31:07 AM node: debug 1
sensor-iot : msg.payload : string[141]
"kampus/labiot/telemetri : msg.payload : Object node_id: "ESP8266_Local_Nod..."
EXPLORER
Search flows
Flows
Flow 1
Flow 2
Subflows
Global Configuration Nodes
PALETTE
mqt
network
mqtt in
mqtt out
connected
Type here to search
29°C
9:38 AM
10/26/2026


Node-RED : Flow 2
localhost:1880/#flow/651414335573690a
Node-RED
Flow 1
Flow 2
Deploy
Edit mqtt out node
Delete
Cancel
Done
Properties
Server
sensor-iot
Topic
kampus/labiot/kendali
QoS
0
Retain
false
Name
Name
Tip: Leave topic, qos or retain blank if you want to set them via msg properties.
Enabled
INFORMATION
kampus/labiot/kendali
Node
"f49b1fbbc8b191eb"
Type
mqtt out
show more
Hold down ⇧ when you click on a node to also select all of its connected nodes
DEBUG MESSAGES
10/26/2026, 9:18:14 AM node: 46410c694d003f00
msg : string[28]
"missing broker configuration"
10/26/2026, 9:31:07 AM node: debug 1
sensor-iot : msg.payload : string[141]
"kampus/labiot/telemetri : msg.payload : Object node_id: "ESP8266_Local_Nod..."
EXPLORER
Search flows
Flows
Flow 1
Flow 2
Subflows
Global Configuration Nodes
PALETTE
mqt
network
mqtt in
mqtt out
connected
Type here to search
29°C
9:38 AM
10/26/2026



Latihan 1 (Analisis Jaringan Internal Docker & Service Discovery):

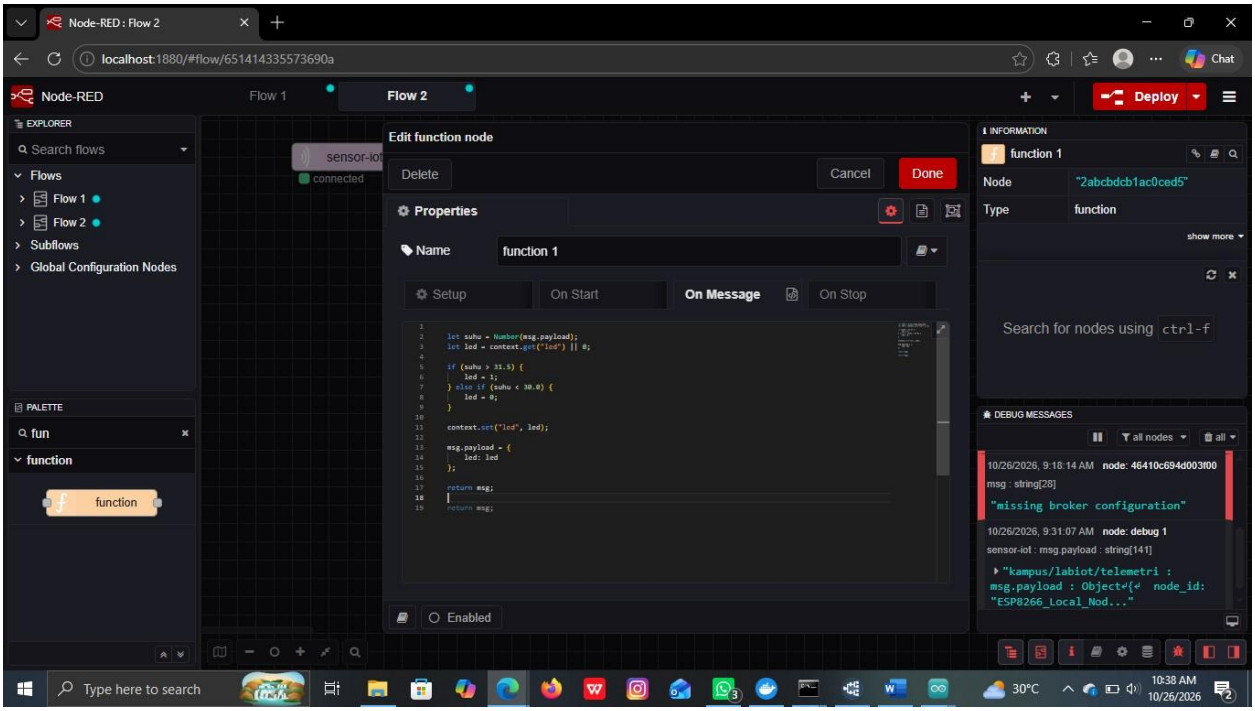
1. Mengapa penggunaan localhost pada Node-RED di dalam kontainer Docker menyebabkan kegagalan koneksi?

Penggunaan localhost pada Node-RED di dalam kontainer Docker akan menyebabkan kegagalan koneksi karena localhost mengacu pada kontainer Node-RED itu sendiri, bukan pada laptop atau kontainer broker MQTT. Ketika Node-RED mencoba terhubung ke localhost:1883, koneksi akan gagal jika tidak ada broker MQTT yang berjalan di dalam kontainer Node-RED tersebut. Oleh karena itu, alamat server broker harus diisi dengan nama layanan mqtt agar Node-RED dapat terhubung ke broker MQTT yang benar.

2. Bagaimana mekanisme Docker Embedded DNS Server dan Bridge Network bekerja?

Docker menggunakan Embedded DNS Server untuk menerjemahkan nama layanan menjadi alamat IP internal kontainer yang sesuai. Ketika kontainer node-red dan mqtt-broker berada pada jaringan Docker yang sama, yaitu Bridge Network yang dibuat oleh Docker Compose, NodeRED dapat menemukan broker menggunakan nama layanan mqtt. Docker kemudian menerjemahkan nama tersebut menjadi alamat IP internal kontainer broker sehingga komunikasi dapat dilakukan melalui port MQTT 1883 tanpa harus menentukan alamat IP secara manual. Mekanisme ini mempermudah komunikasi antarkontainer dan memungkinkan alamat IP berubah tanpa perlu mengubah konfigurasi Node-RED.

Latihan 3 (Implementasi Filter Histeresis pada Rule Engine):



Latihan 4 — Analisis Mekanisme Volume Mount & Persistensi Data 1.
Apa yang terjadi jika baris volume dihapus dari docker-compose.yml?

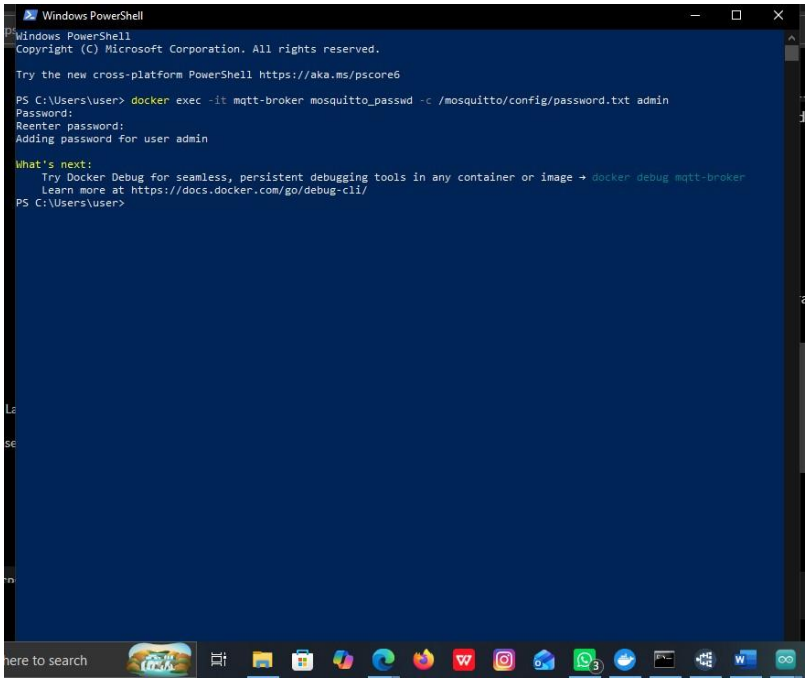
Jika baris volume ./nodered:/data dihapus dari file docker-compose.yml, data alur kerja (flow) dan konfigurasi Node-RED akan disimpan di dalam lapisan penyimpanan kontainer. Apabila kontainer dihentikan dan dihapus menggunakan perintah docker compose down, kemudian dibuat kembali, data yang tersimpan di dalam kontainer lama dapat hilang. Akibatnya, alur kanvas Node-RED yang telah dibuat berisiko hilang dan harus dibuat ulang.

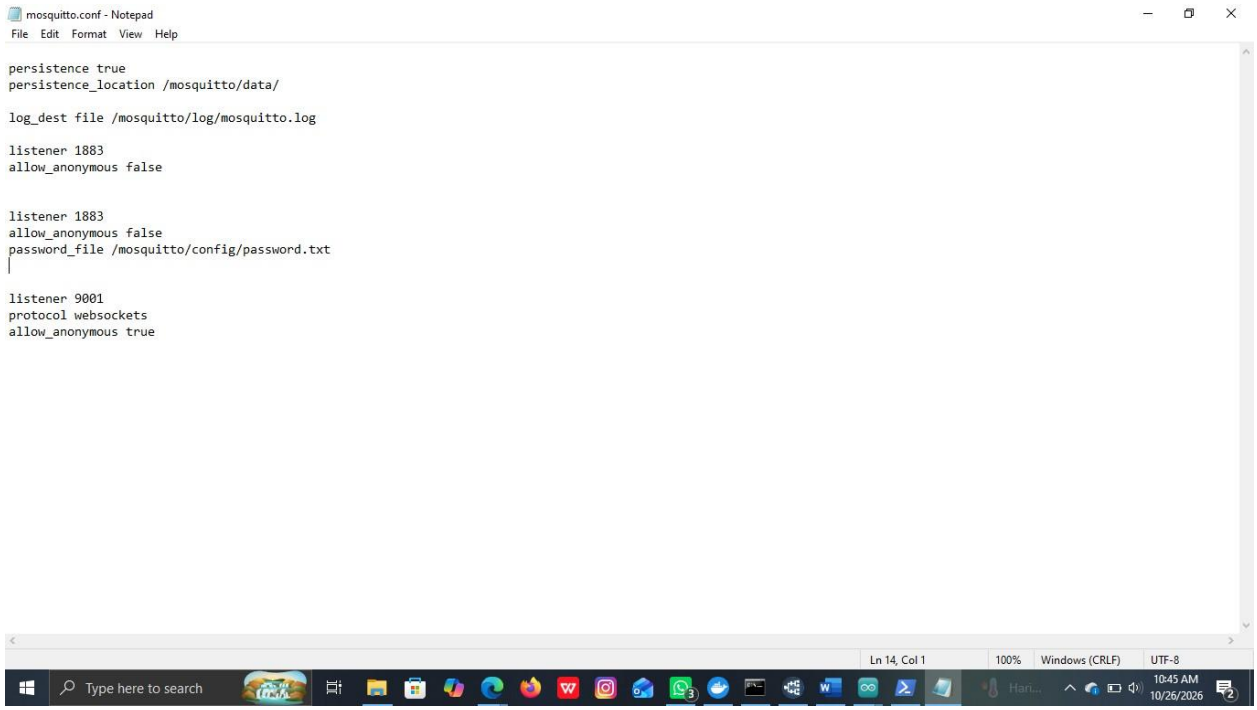
Hal yang sama berlaku pada Mosquitto. Jika pemetaan ./mosquitto/data:/mosquitto/data dihapus, data yang tersimpan di direktori tersebut tidak lagi terhubung dengan direktori host. Data yang hanya tersimpan di dalam kontainer lama dapat hilang ketika kontainer dibuat ulang.

Oleh karena itu, volume mount diperlukan untuk menjaga persistensi data agar tetap tersedia meskipun kontainer dihapus dan dibuat kembali.

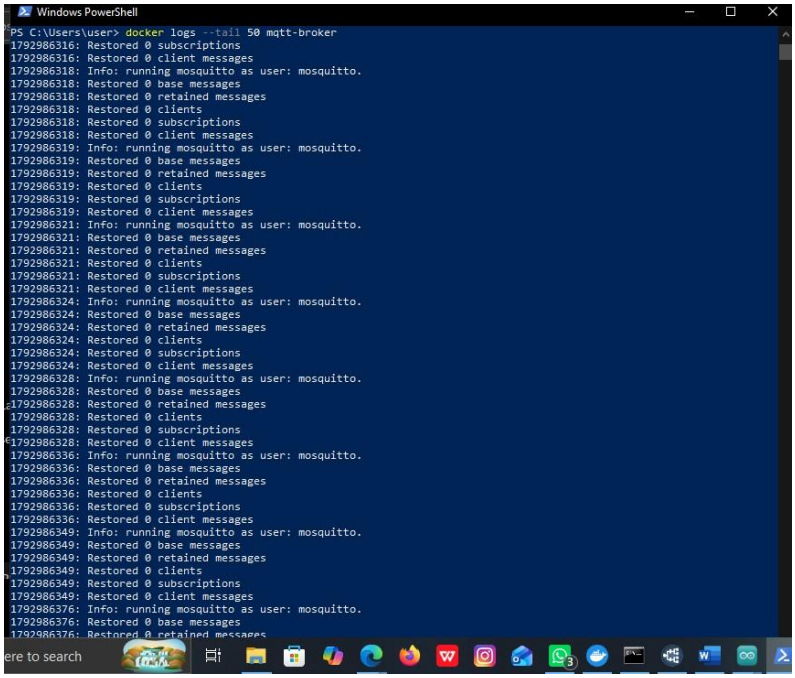
2. Bind Mount menghubungkan direktori tertentu pada komputer host dengan direktori di dalam kontainer, sedangkan Named Volume menggunakan penyimpanan yang dikelola oleh Docker. Keduanya dapat mempertahankan data ketika kontainer dibuat ulang, selama volume atau direktori penyimpanannya tidak dihapus dan pemetaannya tetap dikonfigurasi.

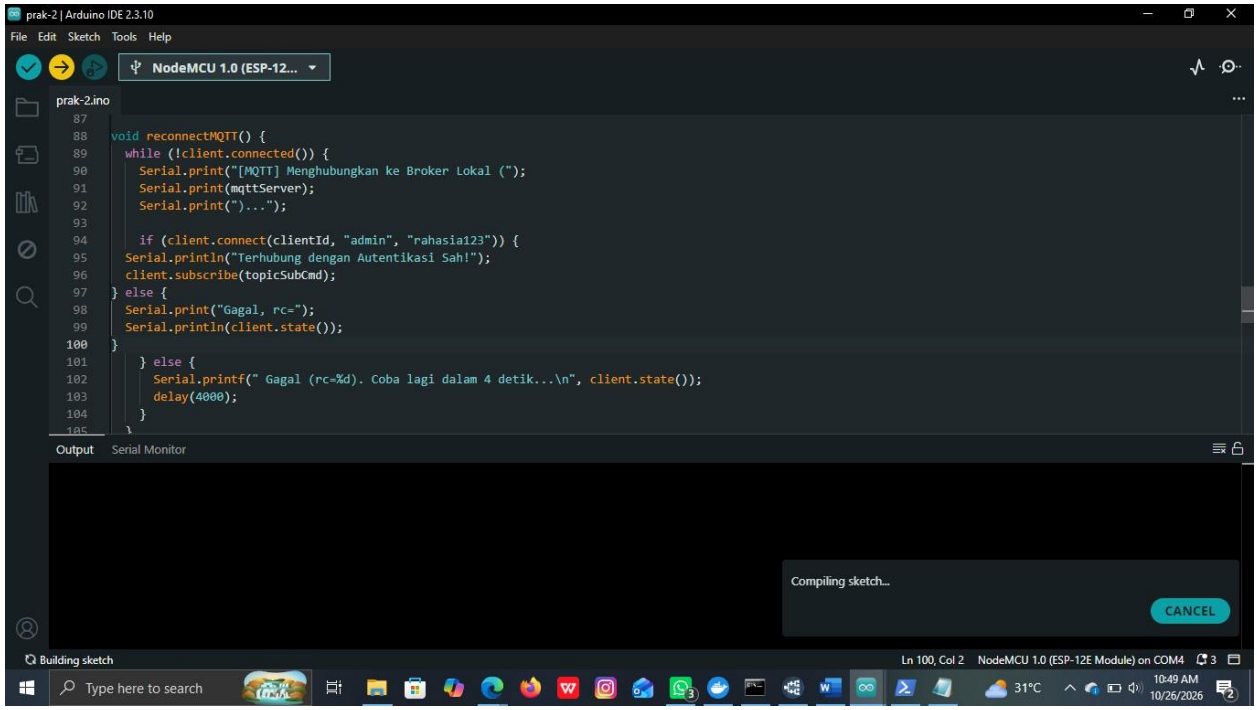
TUGAS

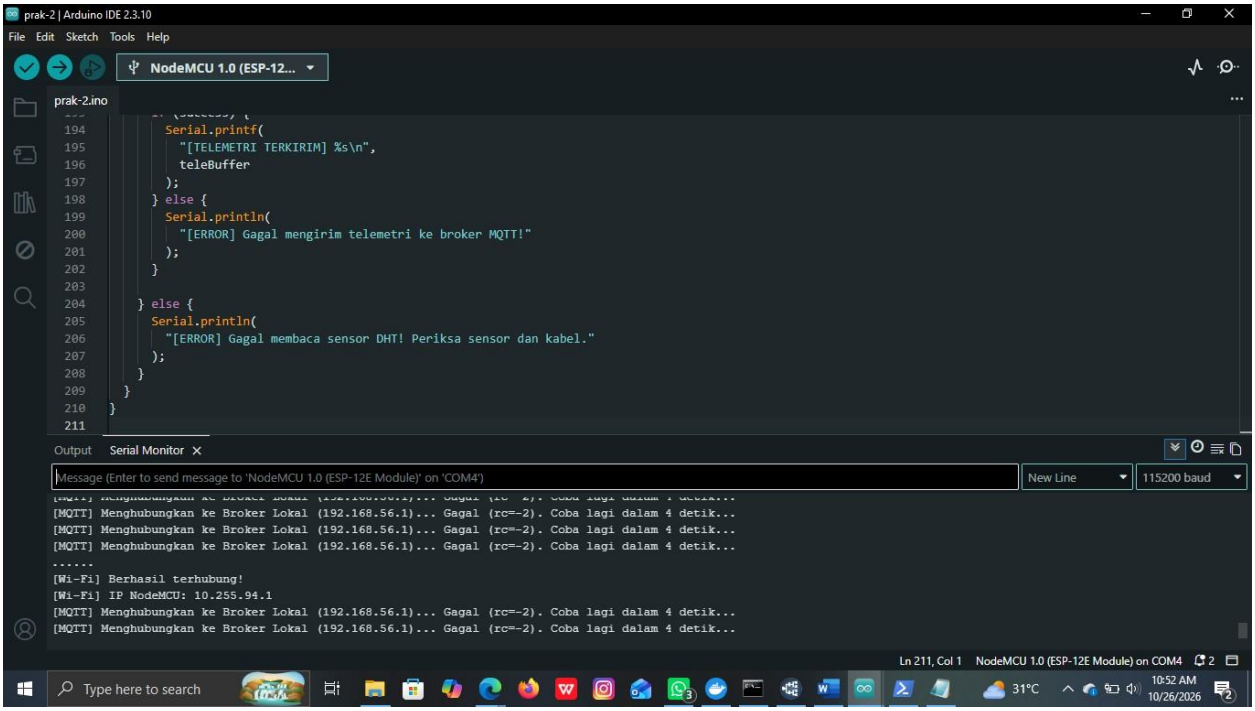

Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Users\user> docker exec -it mqtt-broker mosquitto_passwd -c /mosquitto/config/password.txt admin
Password:
Reenter password:
Adding password for user admin
What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug mqtt-broker
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\user>
here to search

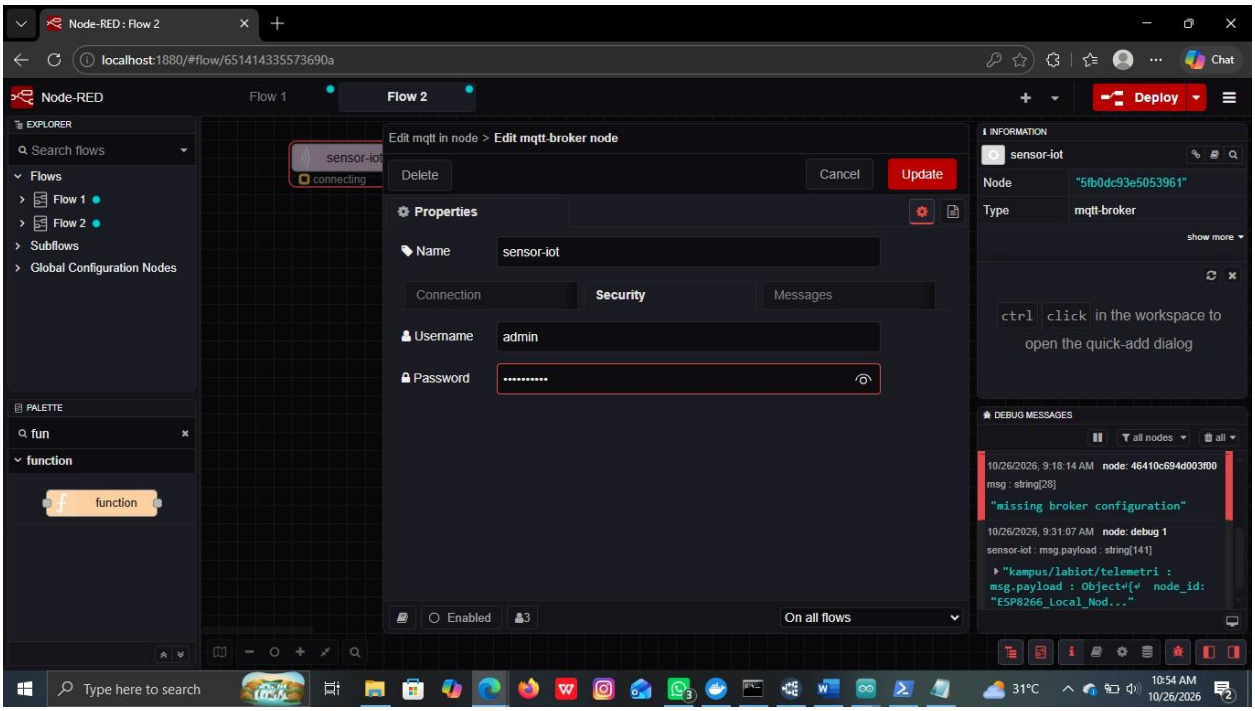

mosquitto.conf - Notepad
File Edit Format View Help
persistence true
persistence_location /mosquitto/data/
log_dest file /mosquitto/log/mosquitto.log
listener 1883
allow_anonymous false
listener 1883
allow_anonymous false
password_file /mosquitto/config/password.txt
listener 9001
protocol websockets
allow_anonymous true
Ln 14, Col 1
100%
Windows (CRLF)
UTF-8
Type here to search
10:45 AM
10/26/2026

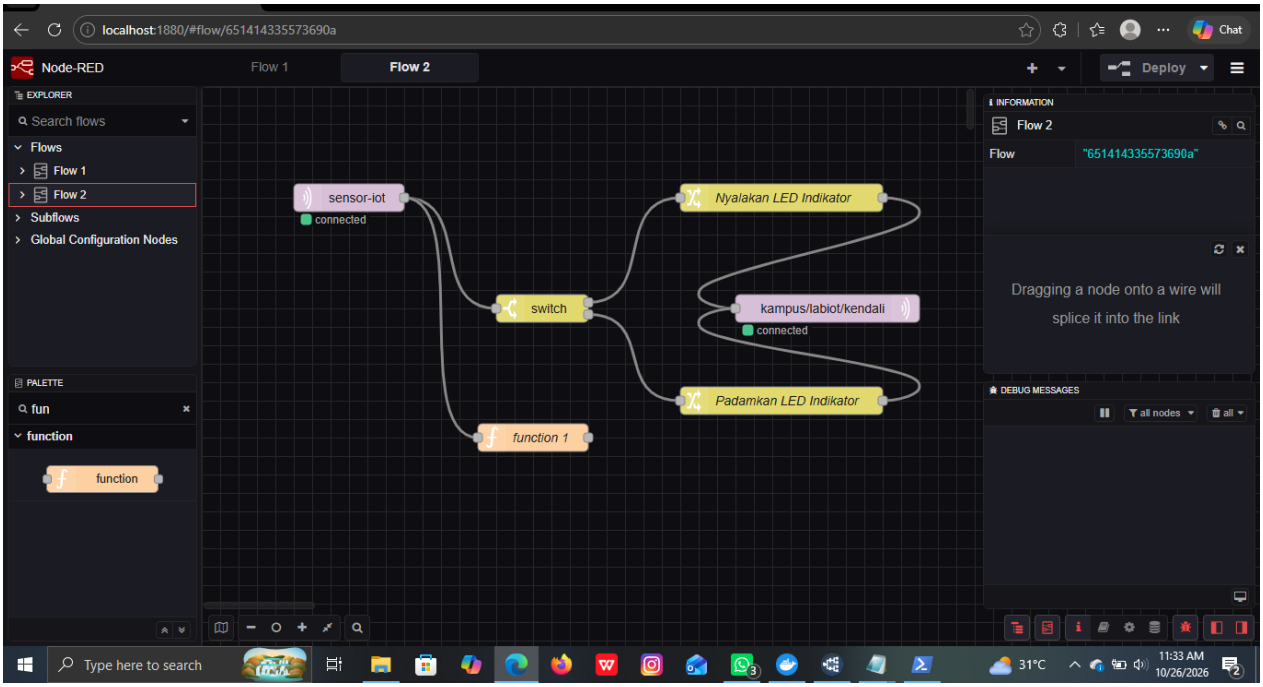
```
Learn more at https://docs.docker.com/go/debug-cli
PS C:\Users\user> docker restart mqtt-broker
mqtt-broker
PS C:\Users\user>
```

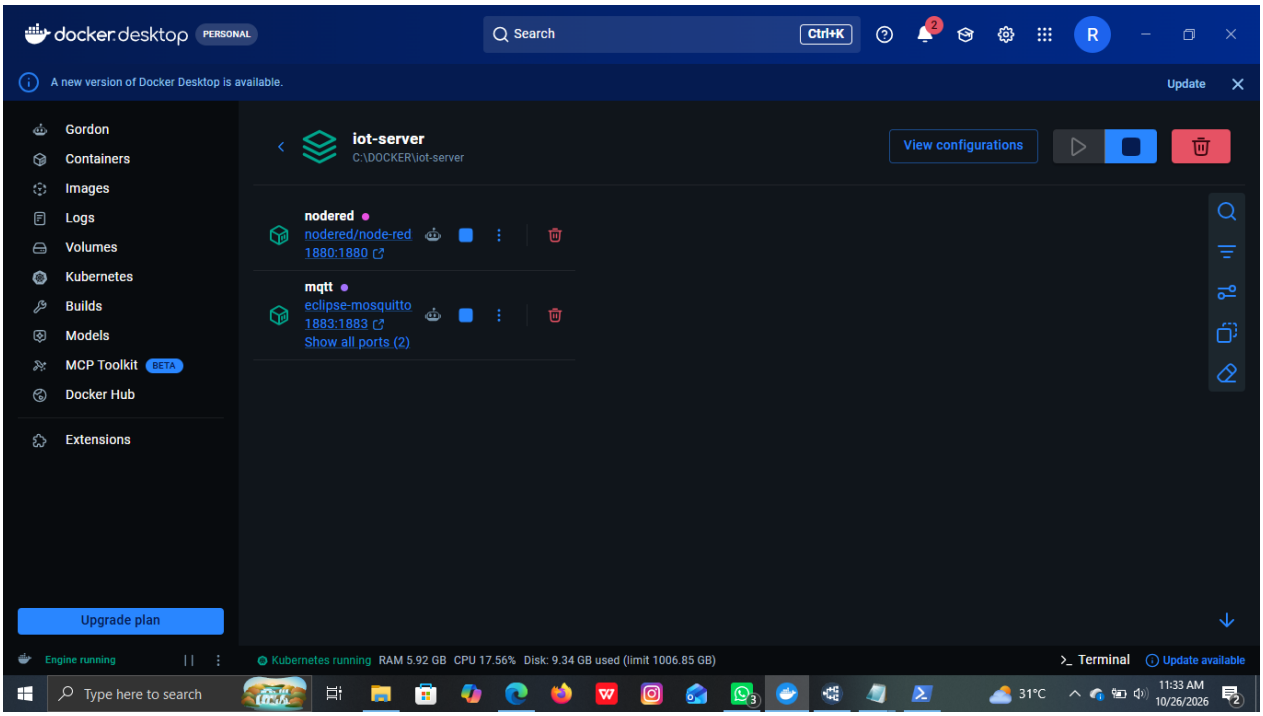





prak-2 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
prak-2.ino
Serial.printf(
"[TELEMETRI TERKIRIM] %s\n",
teleBuffer
);
} else {
Serial.println(
"[ERROR] Gagal mengirim telemetri ke broker MQTT!"
);
}
} else {
Serial.println(
"[ERROR] Gagal membaca sensor DHT! Periksa sensor dan kabel."
);
}
}
}
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM4')
New Line
115200 baud
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
.....
[Wi-Fi] Berhasil terhubung!
[Wi-Fi] IP NodeMCU: 10.255.94.1
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.56.1)... Gagal (rc=-2). Coba lagi dalam 4 detik...
Ln 211, Col 1 NodeMCU 1.0 (ESP-12E Module) on COM4
Type here to search
31°C
10:52 AM
10/26/2026


Node-RED : Flow 2
localhost:1880/#flow/651414335573690a
Node-RED
Flow 1
Flow 2
Deploy
EXPLORER
Search flows
Flows
Flow 1
Flow 2
Subflows
Global Configuration Nodes
sensor-iot
connecting
Edit mqtt in node > Edit mqtt-broker node
Delete
Cancel
Update
Properties
Name
sensor-iot
Connection
Security
Messages
Username
admin
Password
INFORMATION
sensor-iot
Node
"5fb0dc93e5053961"
Type
mqtt-broker
show more
ctrl click in the workspace to open the quick-add dialog
PALETTE
fun
function
function
DEBUG MESSAGES
10/26/2026, 9:18:14 AM node: 46410c694d003f00
msg : string[28]
"missing broker configuration"
10/26/2026, 9:31:07 AM node: debug 1
sensor-iot : msg.payload : string[141]
"kampus/labiot/telemetri : msg.payload : Object{ node_id: "ESP8266_Local_Nod..."
Enabled
On all flows
Type here to search
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10:54 AM
10/26/2026


localhost:1880/#flow/651414335573690a
Node-RED
Flow 1
Flow 2
Deploy
EXPLORER
Search flows
Flows
Flow 1
Flow 2
Subflows
Global Configuration Nodes
sensor-iot
connected
Nyalakan LED Indikator
switch
kampus/labiot/kendali
connected
Padamkan LED Indikator
function 1
INFORMATION
Flow 2
Flow
"651414335573690a"
Dragging a node onto a wire will splice it into the link
DEBUG MESSAGES
all nodes
all
PALETTE
fun
function
function
Type here to search
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11:33 AM
10/26/2026


docker desktop
PERSONAL
Search
Ctrl+K
A new version of Docker Desktop is available.
Update
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Upgrade plan
iot-server
C:\DOCKER\iot-server
View configurations
nodered
nodered/node-red
1880:1880
mqtt
eclipse-mosquitto
1883:1883
Show all ports (2)
Engine running
Kubernetes running
RAM 5.92 GB
CPU 17.56%
Disk: 9.34 GB used (limit 1006.85 GB)
Terminal
Update available
Type here to search
31°C
11:33 AM
10/26/2026